Near-Lossless Long Context under Fixed VRAM via Critical-Span Retention

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Abstract

Long-context inference is limited by KV-cache memory. Training-free eviction methods shrink the cache, but it is often unclear *which* tokens are necessary for near–full-KV quality, and whether online streaming fails from insufficient peak budget or from weak **query-unknown discovery**. On a multi-seed controlled retrieval suite (Qwen3-4B, RTX 3090), we show:

1. **Structure (H1′).** Retaining critical fact tokens plus a small local radius $R^* = 1$ (with attention sinks and a recent/question window) is **necessary and sufficient** for $\varepsilon \approx 0$ single-fact recall: oracle **15/15**, anti-oracle **0/15**, full KV **15/15** (5 seeds $\times$ 3 depths @4k).
2. **Discovery gap.** Online streaming at moderate peak fails under multi-seed hay not because the budget is too small, but because discovery is weak: attention stream@512 is **33%**, while perfect online pin of critical$\pm R$ is **15/15** at the same peak.
3. **Closing the gap (suite).** A **surface novelty detector** (within-prefix rarity + digit/ID-like cues; no final question) reaches $\sim$**93%** multi-seed @4k; with **sticky pin packing** it holds **9/9** cells (3 seeds $\times$ 3 depths) through **40k** at stream@512—peak cache $\sim$**1k tokens**, flat in L.
4. **Public long-doc QA (decomposition).** On a 60-item LongBench slice, sticky novelty lags full ($\sim$0.64$\times$ F1 at peak $\sim$1k). A **posthoc query-aware upper bound** (full prefill $\rightarrow$ compress) recovers **0.95–1.0**$\times$ full F1 at final $B \in \{512, 1024\}$, proving discovery *can* work when the question is known. **Query-hold** streaming trades peak for quality, best $\approx$**0.92**$\times$ full F1 at peak $\sim$2.5k.

The critical-span mechanism transfers across Qwen2.5, Llama-3.2, and hybrid Gemma-4 (full layers). We do **not** claim general long-context SOTA; see Section 6.

1 Introduction

KV cache memory grows with context length L. On a fixed workstation (here: 24 GB), full-cache decode becomes the bottleneck long before "the model cannot attend." Training-free compressors reduce tokens kept, but practice often optimizes average scores rather than **guaranteeing** the spans that encode answers—and online pipelines must score tokens **before** the user question exists.

We study **near-lossless** compression on retrieval-critical prompts: quality Q should match full KV at $\varepsilon \approx 0$ under a multi-seed protocol. The systems objective is

$$L_\varepsilon \;=\; \max\big\{\, L : Q(M,L) \ge (1-\varepsilon)\, Q(\mathrm{Full},L) \,\big\} \tag{1}$$

under **peak** cache / VRAM constraints—not only final decode size after a full prefill.

Claim. Near-lossless training-free KV compression is better understood as **retaining critical local neighborhoods** and **discovering them online before the question exists** than as uniform thinning or posthoc score ranking alone. On open long-document QA, the residual gap is largely an **online retention / peak-cache Pareto**, not a failure of query-aware scoring once the question is available.

2 Related work

Training-free KV eviction and selection. A large line of work compresses the KV cache without fine-tuning by dropping low-scoring tokens. **H2O** [1] keeps heavy hitters under accumulated attention; **StreamingLLM** [2] observes attention sinks and retains initial tokens plus a sliding recent window; **Scissorhands** [3] and **TOVA** [4] prune by recency/attention proxies; **SnapKV** [5] and pyramid-style keepers [6] select important prefix tokens using observation-window attention, often with clustering or hierarchical budgets. Query-aware vs query-agnostic selection (e.g. adaptive budgets conditioned on the prompt) is an active axis; our online setting is deliberately **query-unknown until the question arrives**. These methods primarily address *which scores to use when queries (or an observation window) exist*. Our contribution is complementary: multi-seed **kill tests** that isolate *necessary keep sets*, an explicit **query-unknown discovery gap** for online streaming, and a lightweight surface detector + sticky packing that closes that gap on retrieval suites at flat peak cache.

Streaming and long-context systems. Chunked prefill and stream compression cap **peak** memory rather than only final decode size. We report peak cache tokens, peak VRAM, and prefill latency on a consumer GPU, in the spirit of systems-oriented long-context work rather than leaderboard-only evaluation. Hybrid architectures (sliding-window + full-attention layers; e.g. Gemma-4 class models) still leave full-layer KV as the dominant compressible state; we compress full layers only and keep bookkeeping for hybrid score passes. Retrieval-style KV / page caches (InfLLM-class systems) are complementary infrastructure; we study *which* tokens to retain under a fixed peak budget on a single GPU.

Quantization. KV quantization (e.g. **KIVI** [7]) reduces bytes per retained token and is complementary to eviction. We use fake-int8 only for logical byte accounting in some benches; quality results use bf16 caches unless noted.

Benchmarks. Needle-in-a-haystack and multi-hop synthetic suites stress retrieval under controlled depth/seed. **LongBench** [8] and related public suites (RULER, InfiniteBench) stress general long-document and multi-doc QA. We use a **fixed 60-item LongBench slice** (not a full leaderboard) to separate suite success from open QA, and to measure posthoc upper bounds and peak/quality Pareto curves. We do **not** report full RULER/InfiniteBench leaderboards.

Positioning. We are not proposing a new attention kernel or claiming SnapKV-beating LongBench SOTA. The paper's contribution is a **mechanism + discovery diagnosis + workstation L_ε operating point**, with honest public-data limits and a quantitative online Pareto.

3 Methods

Task (suite). Needle-in-haystack style secrets in seeded filler; depths $\{0, 0.5, 1\}$; success = exact key recall (greedy decode). Stresses: NL facts, multi-needle, multi-hop, prose soft facts, multidoc. Multi-seed protocols: 5×3 at 4k; 3×3 at long L.

H1′ keep set.

$$K^\star \;=\; \text{sinks} \,\cup\, \text{critical} \pm R \,\cup\, \text{recent/question}.$$

Kill arms: full, oracle_r1 ($K^\star$), anti_oracle (drop critical$\pm R$). $R^* = 1$ is the measured default radius.

Posthoc scorer (`seed_valley`). After full prefill of length T, score the last observation window of size W (default W=128, covering the question when present) with shared attention votes over the prefix. Select seed peaks, grow contiguous valleys while score $\geq 0.25\times$ seed, expand $\pm R$, force-keep sinks and the last W tokens, pack to budget B. Used for suite tax tables and as the LongBench **posthoc upper bound** (full prefill $\rightarrow$ compress@B; peak cache $= T$).

Streaming. Chunked prefill (chunk size 512); whenever cache exceeds the active budget, compress online. Peak cache $\approx$ budget + one chunk. Absolute RoPE indices continue from original T at decode time after compress.

Oracle-online upper bound. Force-keep critical$\pm R$ whenever still in cache (perfect discovery) at the same peak class as stream@512.

Surface novelty (query-unknown). For each prefix position i before the final question exists, score token i from within-prefix rarity / first occurrence plus surface cues (digits, ID-like pieces). Take local peaks, expand $\pm R$ into pin sets. **Sticky** registry: once a pin is discovered it remains until packing must drop it; packing ranks pins by score under budget with sinks + recent reserved. This is a lightweight detector, not a learned importance head.

Query-hold (systems lever). Mid-stream: hybrid pins (novelty $\cup$ attention peaks) under `hold_budget`. Near end / final pass: query-aware re-rank and tighten to `final_budget` once the question is in the observation window. Peak $\approx$ hold + chunk.

Decode after compress. Greedy decode uses absolute `next_position=` T. After any final compress we **recompute first-token logits** under the compressed cache (re-forward the last prompt token) so the first generated token is not scored from a stale full/hold cache.

Public LongBench slice. THUDM/LongBench: first 20 items each of `multifieldqa_en`, `qasper`, `hotpotqa` (60 total); contexts truncated to max 4096 tokens (head+tail); token F1 vs gold + substring hit; greedy, max_new=64; same items for all arms. Absolute full F1 is modest under 4B/greedy/truncate—we report **ratios to same-item full**.

Suite-alignment risk. Surface novelty is favored when secrets are surface-distinct against repetitive filler. We report prose/NL/adversarial stresses and public LongBench as honesty checks; LongBench is where novelty *fails* to match full at peak$\sim$1k.

Default API. `prefill_auto(..., mode="stream", discovery="novelty")`.
Use `discovery="query_hold"` for the peak/quality Pareto path.

4 Results

4.1 Figure 1 — Story in four panels

A. H1$'$ kill (4k, 5×3 multi-seed). Full and oracle crit$\pm$1 both **15/15**; anti-oracle **0/15**. Bare critical tokens without radius often corrupt trailing digits (H1 needs local context); $R^* = 1$ restores $\varepsilon \approx 0$. Mean $|\text{oracle}| \approx$ **149** tokens vs full $\sim$4k.

B. Discovery gap (stream@512). Attention valley **33%**; surface novelty **93%** (14/15); oracle pin **100%**. Peak budget is in the same class ($\sim$1k with chunk). Failures of valley at 512 are **detector** failures, not "stream cannot work."

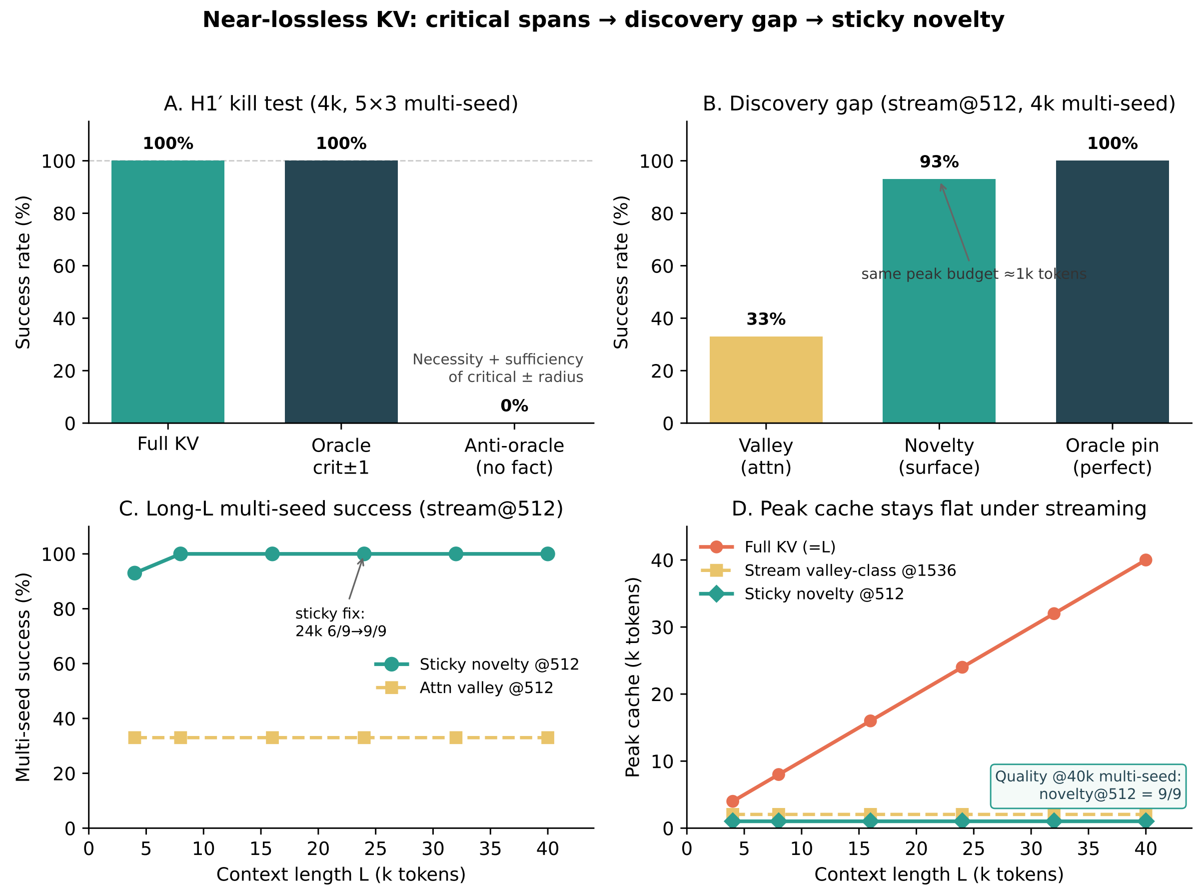

Figure 1: **Story figure.** **(A)** H1′ kill test @4k multi-seed (5×3): full and oracle crit±1 both 15/15; anti-oracle 0/15. **(B)** Discovery gap at stream@512: valley 33%, surface novelty 93%, oracle pin 100%. **(C)** Long-L multi-seed success: sticky novelty@512 holds 9/9 through 40k; valley@512 stays end-only ($\sim$33%). **(D)** Peak cache tokens stay flat ($\sim$1k) under sticky novelty stream@512 while full KV grows with L.

C. Long L multi-seed (3×3). Sticky novelty@512 is **9/9** at 16k, 24k, 32k, and **40k**. Non-sticky novelty degraded at 24k ($\sim$6/9) via re-rank thrash; sticky packing fixed it. Valley@512 remains end-depth-only ($\sim$33%) under multi-seed hay as L grows.

D. Peak cache. Full KV scales with L; sticky novelty stream@512 stays $\sim$**1k** tokens. Prior valley operating points often used $\sim$1.5–2k peak for long L.

4.2 Scorer tax (posthoc, question known)

Table 1: Posthoc budget tax vs oracle keep set (4k multi-seed).

Method	All-cell $B_{\min}$	Tax vs mean |oracle|
oracle_r1	$\sim$149	1.0×
seed_valley (multi-seed all-cell)	**192**	$\sim$**1.29**× / mean cell tax $\sim$**1.16**×
SnapKV (single-needle mid, same scorer suite)	192	$\sim$1.24× vs oracle mid
seed_valley (single-needle mid)	**176**	$\sim$**1.14**× vs oracle mid

Posthoc is near-oracle-tight on the multi-seed suite; SnapKV is competitive but seed_valley is slightly tighter at mid-depth single-needle (FINDINGS scorer budget). **Stream is not** near-oracle until discovery improves.

4.3 Stress and transfer (summary)

Table 2: Stress suite and cross-model transfer (summary).

Check	Result
NL facts (names/places) sticky novelty@512	**15/15** multi-seed
Code needles sticky	**15/15**
Adversarial ID-like filler (pre-sticky)	novelty **100%** vs valley **33%**
Multi-needle recall_all @512	novelty **5/5** (`max_new`$\geq$96); valley **0/5**
2-hop (Alice$\to$id$\to$password) @512	novelty **5/5**; valley **1/5** (needs @1024 for 5/5)
3-hop + distractors @512	novelty **5/5**; valley **0/5** (needs @1024 for 5/5)
Prose soft fact (no digits) @512	novelty **15/15**; valley 53%
Multidoc (6 titled docs) @512	novelty **15/15**; valley/oracle_pin **5/15**
External-style mixed QA (10$\times\sim$4k)	novelty **10/10** hits (= full); valley **0/10**
Gemma-4 E4B hybrid novelty@512	multi-seed **9/9** @4k (valley needs $\sim$1024)
Qwen2.5 / Llama-3.2	H1 holds; family-specific posthoc floors

4.4 Public LongBench decomposition

Synthetic suite success does not automatically transfer to open long-document QA. On the fixed 60-item slice (same items for all arms), online novelty stream@512 is far below full ($\sim$0.64$\times$ F1 at peak $\sim$1k; Table 4), and only slightly above attention streaming on an earlier same-slice run (FINDINGS: valley F1 0.18 vs novelty 0.19). A posthoc query-aware upper bound—full prefill, then `seed_valley` compress with the question in the observation window—recovers **0.95**$\times$ full mean F1 at final B=512 and $\sim$**1.0**$\times$ at 1024/2048 (Table 3). Peak remains full L for posthoc; the point is diagnostic: **query-aware scoring can keep answer-bearing tokens** when allowed to see the full prefix and the question.

Online paths must drop tokens before the question exists. Table 4 reports a **query-hold Pareto**: larger mid-stream hold improves recovery after query-aware tighten; final budget 1024 consistently beats 512 at fixed hold. Best measured online point: hold 2048 $\to$ final 1024 $\approx$ **0.92**$\times$ full F1 at mean peak $\sim$2.5k tokens—between novelty@1k (0.64$\times$) and full@L.

Interpretation. Open long-doc QA on this slice is not primarily "seed_valley cannot find answers when query-aware." It is an **online retention** problem. Systems claims should be stated as a peak/quality curve, not as free near-full quality at peak$\sim$1k outside retrieval suites.

4.5 Systems resources (peak VRAM / latency)

Mid-depth needle, seed 0. Novelty stream@512 keeps peak VRAM and decode KV essentially flat:

Peak cache under novelty stays $\sim$1024 tokens at all three lengths.

5 Adaptive policy (systems)

`prefill_auto` applies L-based budgets + per-model floors. With default `discovery="novelty"`, single-needle stream@**512** is the measured multi-seed operating point through **40k**. Attn-only discovery still needs larger floors ($\sim$1536 multi-seed @4k). Hybrid Gemma: compress **full** layers only; sliding layers keep absolute prefix bookkeeping for score-pass.

Table 3: LongBench slice (60 items @4k truncate): posthoc upper bound vs full. Peak for posthoc is full L (not a free systems win).

Arm	Mean F1	Substr hit	Peak	F1 / full
full	**0.282**	25%	$= L$	1.00×
posthoc@512	0.267	27%	$= L$	**0.95**×
posthoc@1024	0.284	28%	$= L$	∼1.01×
posthoc@2048	0.285	27%	$= L$	∼1.01×

Table 4: LongBench slice: online quality vs peak cache (query-hold Pareto).

Arm	Mean F1	Hit	Mean peak	F1 / full
novelty@512	0.179	10%	1024	0.64×
query_hold h1024→f512	0.199	18%	1536	0.70×
query_hold h1024→f1024	0.231	22%	1536	0.82×
query_hold h2048→f512	0.212	17%	∼2531	0.75×
query_hold h2048→f1024	**0.260**	22%	∼**2531**	**0.92**×
query_hold h4096→f1024	0.249	28%	∼3876	0.88×

6 Limitations and what we do *not* claim

6.1 What we claim

- On this **controlled multi-seed retrieval suite**, critical$\pm R^*$ is necessary/sufficient for $\varepsilon \approx 0$ single-fact recall.
- Stream failures at moderate budget are primarily a **query-unknown discovery** problem (oracle-online upper bound).
- Sticky surface novelty **closes most of that gap** through **40k** multi-seed at peak cache ∼1k on the primary model, with transfer smoke to other small instruct models including hybrid Gemma-4.
- On a fixed public LongBench slice, **posthoc query-aware compression reaches 0.95× full F1** at final B=512 (peak=L), while online novelty lags (∼0.64× at peak∼1k); **query_hold** recovers a measured Pareto (best ∼0.92× full F1 at peak ∼2.5k). Absolute full F1 is modest under 4B/greedy/4k truncate.

Table 5: Peak resources: full chunked prefill vs sticky novelty stream@512.

L	full VRAM	nov. VRAM	full KV	nov. KV	nov. prefill
4k	8.5 GB	**8.3 GB**	569 MB	**72 MB**	1.6 s
16k	10.6 GB	**8.3 GB**	2296 MB	**72 MB**	5.9 s
40k	14.5 GB	**8.3 GB**	5754 MB	**72 MB**	17.6 s

Table 6: Scope boundaries (honest).

Not claimed	**Why**
General long-context SOTA	No full RULER / LongBench / InfiniteBench leaderboard evaluation.
All task types	Primary suite is retrieval / needle-class (+ multi-needle, hops, prose, multidoc stress). Full reasoning chains, code repos, and full LongBench/RULER leaderboards untested.
Oracle-tight online budgets for free	Sticky novelty multi3/hop is **5/5**@512 with adequate decode budget; multi-secret packing still stresses oracle_pin@512 (multi3 **2/5**). Suite-alignment of surface novelty remains.
Production memory stack	Fake-int8 is logical accounting only; no fused kernels or serving productization.
Novelty as universal "importance"	Exploits surface distinctness vs repetitive filler; ID-like filler / filler-like secrets can still confuse discovery.
Large models / training-free SOTA	Primary evidence is $\sim$3–4B instruct models on one GPU class.
Statistical finality	Multi-seed N modest (5×3 @4k; 3×3 long-L). Lab-grade, not a large meta-analysis.
$\varepsilon = 0$ beyond measured envelope	Sticky multi-seed measured through **40k**; longer L is extrapolation.
Peak VRAM = final decode KV	Streaming caps **peak cache tokens**; weights and activations still dominate total VRAM.

6.2 Threats to validity

- **Greedy decode** and chat templates may interact with exact-match / token-F1 scoring; LongBench absolute full F1 is low ($\sim$0.28).
- **Filler distribution** is synthetic; surface novelty is suite-aligned to distinct secrets in repetitive hay—LongBench is the external honesty check.
- **Multidoc packing:** novelty@512 can beat labeled oracle_pin@512 (15/15 vs 5/15), so keep-set definitions and packing under multi-span budgets remain open.
- **Hybrid models:** only full layers are compressed; sliding-window behavior is infrastructure-sensitive.
- **Sample size:** multi-seed N is lab-grade (5×3 @4k; 3×3 long-L); "9/9 through 40k" is cells on this protocol, not a large meta-analysis.
- **Selection bias toward positive arms:** we iterated methods until discovery worked; negative results (scored capsules alone, non-sticky long-L) are reported but the path is research-in-the-loop.
- **Community baselines:** stream tables emphasize valley / novelty / oracle / query_hold; SnapKV is reported posthoc on the scorer-tax suite, not as a full multi-seed stream LongBench arm.

7 Conclusion

Near-lossless training-free KV compression on retrieval tasks is structured by **critical local neighborhoods**. When the question is known, a simple attention valley scorer approximates the oracle keep set with small tax—and on a public LongBench slice, posthoc query-aware compression reaches **0.95**$\times$ full F1 at final B=512 (peak=L). When the question is **not** known—as in online streaming—the binding constraint is **discovery**/retention before the query. Sticky surface novelty is a lightweight query-unknown detector that, on our multi-seed suite, restores high success through **40k** context at a **flat $\sim$1k-token peak cache** (9/9 cells on the long-L protocol). On open long-doc QA, quality becomes a **Pareto** in peak cache (query-hold $\sim$0.92$\times$ full F1 at $\sim$2.5k peak). Honest next steps are better mid-stream pins, multi-entity packing, and head-to-head stream baselines against published keepers—not larger single-needle grids alone.

Reproducibility

```
# Multi-seed H1 + scorer tax
python experiments/bench_paper_rigor.py --seeds 0,1,2,3,4 --ctx 4096

# Discovery gap / novelty
python experiments/bench_capsules.py --novelty --oracle-online \
  --skip-capsules --budgets 512
python experiments/bench_novelty_stress.py
python experiments/bench_novelty_longL.py \
  --ctx 16384,24576,32768,40960 --arms novelty --budgets 512

# LongBench posthoc UB + query_hold Pareto (60 items)
python experiments/bench_external_slice.py --source longbench --n 60 \
  --arms full,posthoc --budgets 512,1024,2048
python experiments/bench_external_slice.py --source longbench --n 60 \
  --arms full,novelty,query_hold --budgets 512 \
  --hold-budgets 1024,2048,4096 --final-budgets 512,1024
```

```
# Figure + transfer
python experiments/plot_paper_figures.py
python experiments/bench_transfer.py --model google/gemma-4-E4B-it
```

Primary model: `Qwen/Qwen3-4B-Instruct-2507`, transformers, CUDA bf16, RTX 3090.
Artifact index: `results/FINDINGS.md`.
Markdown twin: `papers/PAPER_DRAFT.md`.
Build PDF: from `papers/`, run `powershell -File build_pdf.ps1`.

References

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